Sreenithya Mallavarapu

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Education

University of Wisconsin Madison — Madison, Wisconsin
Masters in Biological Systems Engineering — *Sep 2024 – May 2026*

Indian Institute of Technology Jodhpur — Rajasthan, India
Bachelor of Technology in Bioscience and Bioengineering — *Dec 2020 – May 2024*

Relevant Coursework

Biostatistics, Computational and Systems Biology, Biophysics and structural biology, Biochemistry, Introduction to Machine Learning, Deep Learning, Differential Equations, Probability, statistics and stochastic processes

Experience

University of Wisconsin Madison — October 2024 - Present
Graduate Student Researcher — *Madison, WI*

- Investigating the interactions of GABA with vitamins and short-chain fatty acids along the bloodstream and vagal passage, aiming to understand its role in neuronal function.
- Conducting computational screening of molecular interactions, followed by in vitro validation using brain cell lines to assess bioavailability and neuroactive effects.
- Developed expertise in molecular docking, statistical and data analysis for studying neuroactive compounds.

Advanced Centre for Treatment, Research, and Education in Cancer (ACTREC) — June -July 2023
Summer Intern — *Mumbai, India*

- Developed and implemented a machine learning algorithm to predict the pathogenicity of PTEN gene mutations associated with breast cancer.
- Compiled a comprehensive training dataset using computational bioinformatics tools such as PolyPhen-2, SIFT, and Align GVGD, quantifying mutation impact on a 0-1 scale.
- Optimized algorithm performance and validation through Python, Orange Tool, and Microsoft Excel.
- Technical Skills: Python, Machine Learning, Bioinformatics Tools, Data Analysis

Indian Institute of Technology Bombay — May -July 2022
Summer Intern — *Mumbai, India*

- Conducted expression, purification, and crystallization of Topoisomerase (Topo67 WT) protein.
- Gained hands-on experience with advanced biochemical techniques, including SDS-PAGE, FPLC, Ni-NTA chromatography, and UV spectrophotometry.
- Collaborated on X-ray diffraction studies for protein-ligand binding analysis using computational modeling.
- Technical Skills: Protein Purification, Crystallization, Computational Drug Discovery, Spectrophotometry

Projects

Genetic Architecture Analysis Using LD Score Regression | *Graduate Coursework* — Feb 2025 – Present

- Performed genetic correlation and heritability estimation using the LDSC package. Analyzed publicly available GWAS summary statistics for traits including IBD, SCZ, EA, and AF.
- Conducted meta-analysis of GWAS datasets and explored cross-trait genetic correlations.
- Utilized R, bash scripts, and LDSC tools for interpreting outputs (heritability estimates, intercepts, rg values).

Deep Learning-Based Quantification of Tunneling Nanotubes (TNTs) | *B.Tech Project* — Aug 2023 – May 2024

- Developed a deep learning model using YOLO on the COCO dataset to quantify TNTs in brain tumor cells from image datasets (400 brain tumour cells).
- Performed image augmentation in RoboFlow to enhance detection accuracy.
- Implemented compactness and circularity as features for scoring for cell spheroids.

Technical Skills

Languages: Python, R, SQL, C
Databases: NCBI, NGS, Swiss-prot, PDB, UniProt
Developer Tools: VS Code, Colab, Pymol, Pyrx, BLAST, FASTA, Microsoft office, Linux
Libraries: pandas, LDSC, NumPy, Matplotlib